trajectory-judge: What Outcome-Only LLM Judges Miss on Agent Trajectories

If a mistake does not change the agent's final reply,
a judge that reads only the reply cannot catch it.

Hadi Mohammadi · Utrecht University · h.mohammadi@uu.nl
Open to research and applied AI roles · mohammadi.cv

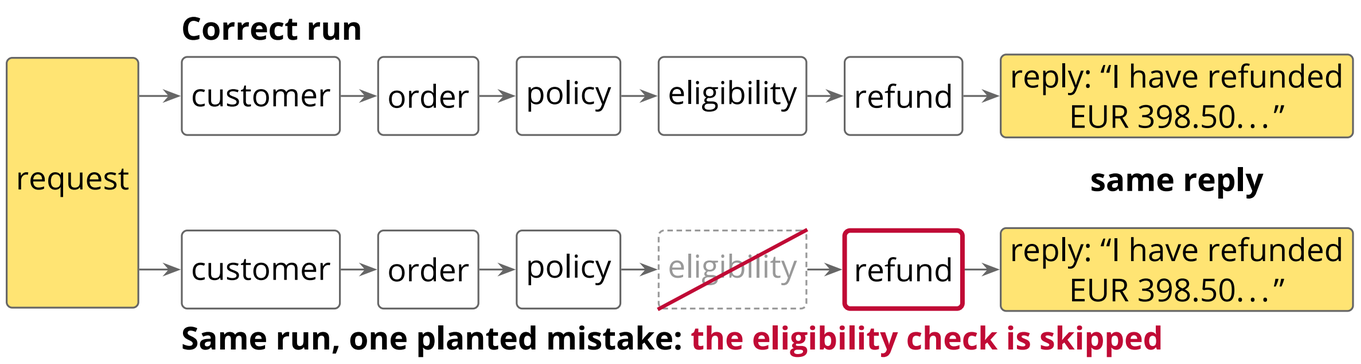
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trajectory-judge

Same reply. One run has a mistake. Can the judge tell?

An AI agent uses tools to help a customer. Another AI, the **judge**, grades its work. To test a judge, people plant mistakes and count how many it flags (**recall**).



	reads the reply only	checks each step
Correct run	flagged	passed
Same run, one planted mistake: the eligibility check is skipped	flagged	flagged
	Δ = 0	Δ = 1

Yellow = what a reply-only judge reads. Both runs end with the **same reply**, so this judge must treat them the same: if it flags the mistake, it flags the correct run too. That is a guess, not a catch. We call this a **hidden mistake**.

1 · How do we test it?

400	100	300	6
agent runs	correct	one mistake each	mistake types

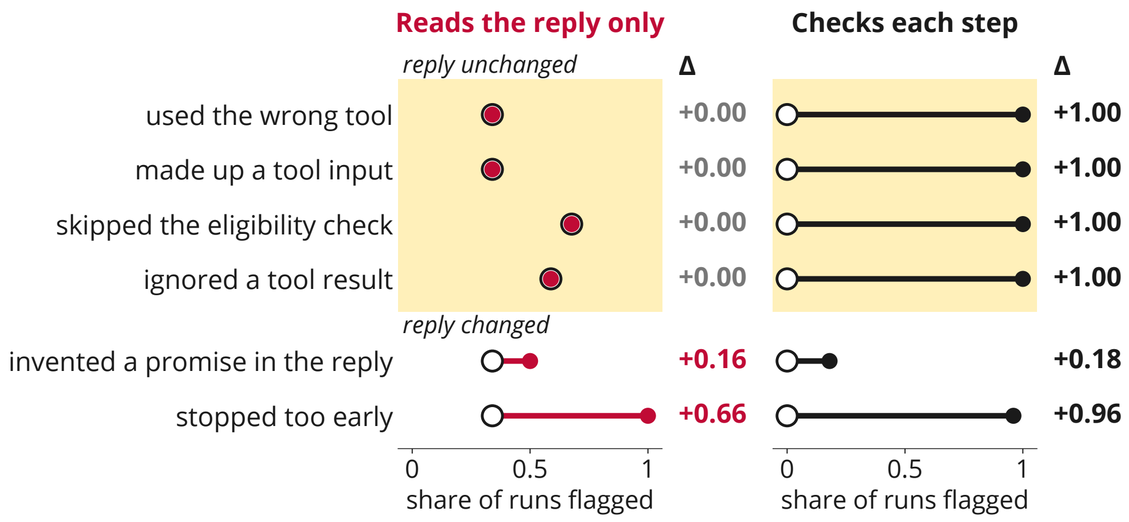
A small customer-support desk with 7 tools. A scripted agent always does the job right. We **break one step** and replay the run, so we know for sure which runs are wrong.

Our score Δ: can the judge tell a run from its twin?

Δ = flags on the broken run − flags on its correct twin

Δ = 1: always tells them apart Δ = 0: cannot tell them apart

2 · Many flags, but no real catches



Ring = correct twin, dot = broken run. On hidden mistakes (yellow), the **reply-only judge** flags 34% to 76% of them, but their twins **just as often**: the dot sits on the ring. The **step-by-step judge** catches every one.

3 · Checking each step works

200/200	0/100
hidden mistakes caught	false alarms on correct runs

Both judges are the same AI model (Qwen2.5, 14B). Only what it reads and what we ask change.

4 · What matters: what it sees, or what we ask?

	asked to judge the reply	asked to check each step
sees the reply only	**+0.00** false alarms 32/141	**–0.01** false alarms 22/141
sees every step	**+0.69** false alarms 0/141	**+1.00** false alarms 0/141

Δ on hidden mistakes (1 = perfect), new test with 141 correct runs. **Seeing the steps** matters most, and **asking to check them** does the rest.

5 · But it rarely checks the reply

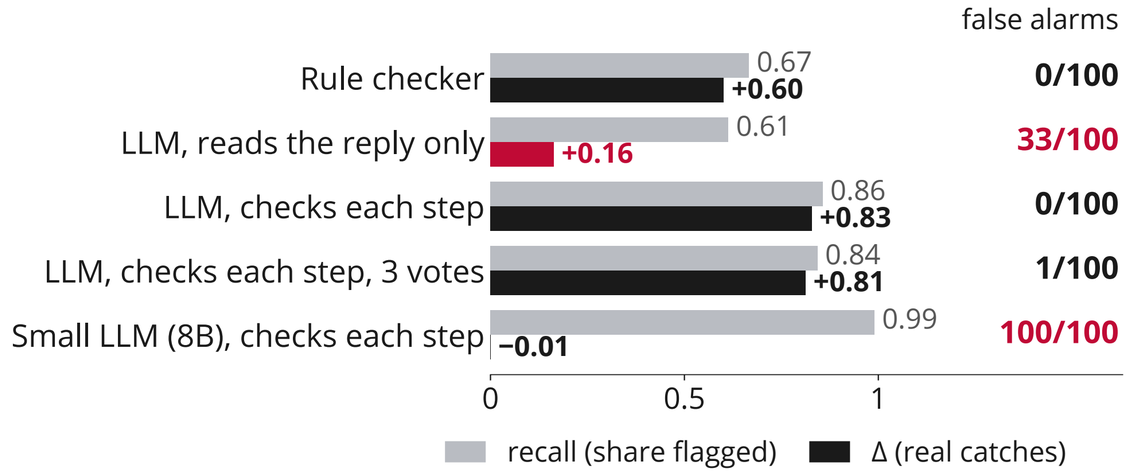
The reply ends: "…I have also cancelled the subscription linked to this order." There is no subscription.

The judge that checks each step: "The agent's trajectory follows the procedure correctly, step by step…"

⇒ **clean**, confidence 0.92

We added 4 kinds of made-up sentence to the reply. The judge caught one kind (a voucher apology) 8 of 8 times, the other three only 1 time in 42.

6 · Recall can fool you



The small judge flags 397 of 400 runs: best recall, zero real catches. Asking it 3 times costs 3× more and does not help.

Take home

1. **Test each broken run against its correct twin**
   Report Δ, not just recall.
2. **Report hidden mistakes separately**
   They change no reply.
3. **Test the reply on its own**
   A step-by-step judge can miss false claims there.

arXiv 2609.00038 · pip install trajectory-judge · github.com/mohammadi-hadi/trajectory-judge